Boping Li

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Education

The University of Texas at Austin — Austin, TX

B.S. in Electrical and Computer Engineering, GPA: 3.8/4 — *Aug. 2024 – Present*

- **Relevant Coursework: Computer Architecture**, **Algorithms**, **Embedded Systems**, **Circuit Theory**, Linear Systems and Signals, Software Design I & II, Random Processes, Discrete Math, Digital Logic Design

Experience

Embedded Software Engineering Intern — Summer 2026

Inspire Semiconductor LLC — *Austin, TX*

- Programmed a **Linux PCIe endpoint function driver** for **RISC-V** with **QEMU** and hardware backends
- Wrote a **U-Boot GPIO driver** via the Driver Model with **devicetree** binding and full **Kconfig** support
- Created a new **Yocto** machine target for **RTL simulation**, validating **SoC firmware** before silicon tape-out

Field Engineer Internship — Summer 2025

SkyStar Storage Energy Service LLC — *Potosi, WI*

- Restored site capacity by **38%** through targeted **firmware updates** and onsite preventative maintenance
- Updated **Compressor Thermal Management Systems (TMS)** firmware to restore inefficient **CATL** units
- Performed maintenance on **128 battery units**, preventing environmental equipment damage onsite at the facility

Student Intern / Researcher — Summer 2021 – 2023

EnPower Resources Inc. — *Sugar Land, TX*

- Fabricated remote **Oil-well Pressure Monitors (OPM)** using microcontrollers and custom **PCB design**
- Leveraged the **Particle Argon** microcontroller's built-in cellular system to connect remotely to the **IoT**
- Installed **OPMs** onto existing MCUs at oil sites to prevent gas leaks and enable wireless connectivity

Outreach and Volunteer Lead — Mar. 2022 – Present

KomoriCon — *Austin, TX*

- Planned and executed an anime convention serving **3,000+ guests**, profiting over **$16k** in a single day
- Managed communications for **213 unique vendors**, **50 volunteers**, **16 panelists**, and venue representatives
- Led day-of **volunteer training** and staff reorganization to cover occupied roles at the convention

Research and Projects

Embedded Connect Four Game | *C++, Assembly, MSPM0* — Feb. 2025 – May 2025

- Developed a **C++** and **Assembly** game with sound effects on the **MSPM0** microcontroller using **OOP**
- Engineered game logic, graphics rendering, **finite state machines**, **DAC**, **ADC**, and I/O systems
- Utilized **UART** serial communication to load the game and sprites onto a graphical **LCD** screen
- Verified **UART**, **DAC**, and **ADC** output on a digital **oscilloscope** to correct data transport issues

FPGA Timer | *RTL Design, Verilog, Xilinx Vivado* — Aug. 2025 – Dec. 2025

- Implemented a multi-mode timer using **RTL-design** methodology on a **Xilinx Basys3 FPGA** board
- Applied a **high-level state machine**, custom **datapath**, and a controller **FSM** to design the timer
- Integrated **10-switch** arrays, button inputs, and four **7-segment displays** with **10ms** resolution
- Operated **Xilinx Vivado** synthesis, implementation, and **bitstream** generation to verify hardware functionality

Garage Controller | *C++, Particle.io, PCB Design* — Summer 2025

- Built a wireless controller to open the garage, turn on/off the lights, and read **temperature** remotely
- Integrated **C++** code with **Particle** functions to interact between microcontroller and mobile app
- Scripted unique **Particle.io** functions to retrieve and process analog temperature data remotely
- Soldered **diodes**, **resistors**, microcontrollers, and two **relays** onto custom-designed **PCBs**

Achievements and Skills

Achievements: AIME Qualifier (2022–23), Houston Half Marathon 2026, GMTK Game Jam 2025 – Rank #4640/9598
Languages: Java, Python, C++, C, JavaScript, HTML5, CSS, Verilog, Matlab
Tools & Platforms: Git, GitHub, Linux, Yocto, Linux Drivers, Vivado, KiCad, CAD, 3D Printing, Soldering
Other: English, Chinese (Mandarin), CPR Certified, Object-Oriented Programming, AI, Leadership, Communication, Teamwork, Collaboration